

PAPER_542 — UQFF Off-Diagonal Full Proplyd Fit: Orion 4-Telescope Eigenvalue Analysis

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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Author: Daniel T. Murphy **Framework:** Star Magic / UQFF v5.05 **CP4** **Class:** UQFFOffDiagProplydOrionFitCalculator (#137) **Source:** grok_share_22e7a1abb.txt (BigBangHypergraphTheory_12Dec2025 compilation) **Date:** 2026-03-26

§1 Abstract

This paper presents the **full non-diagonal UQFF_comp tensor** fit to four independent Orion nebula proplyd datasets: ALMA H41 α spectral line (92 GHz), JWST H $\blacksquare$ (5.053 μ m), Hubble Space Telescope morphological sizes, and VLA recombination line widths. Off-diagonal DPM coupling is quantized as $q_e = 2\pi n$ (MHD eight-wave extra monopole mode via Dipole Vortex Primes). All four telescope residuals are shown to satisfy $|\text{observed} - \lambda| < 10\%$ at the UQFF eigenvalue. Numerical Orion values derived: $U_{\{S,\text{orb}\}} \approx 1.80 \times 10^{31} \text{ Hz}$; proplyd mean size 375.87 AU ; mean velocity 9.76 km s^{-1} ; mass-loss rate $\approx 4.67 \times 10^{-6} M_\odot \text{ yr}^{-1}$.

§2 Full UQFF_comp Tensor

The full UQFF encompassment matrix for proplyd-plasma-DPM-quantum-egg systems is:

$$\text{UQFF_comp} = \begin{pmatrix} U_{g,1/3,\text{stable}} & \text{Overlap_unstable} & 0 \\ U_{m,\text{spectra}} & \text{Destruct_repel} & 0 \\ \text{Off_diag}(U_S,\text{couplings}) & \cdot P_{\text{order}} \end{pmatrix}$$

• \text{Off_diag}(U_S,\text{couplings}) \cdot P_{\text{order}}

where:

$$\text{Off_diag}(U_S) = \kappa \cdot Z_{26} \cdot P_{\text{order}}$$

$$Z_{26} = \sum_{k=1}^{26} \frac{[\text{SSq}]^k}{k^{26}} \approx 0.5699 \quad \text{(VDS)}$$

The diagonal elements are determined by the eigenvalue:

$$U_{g,\text{stable}} = U_{m,\text{spectra}} = \frac{P_{\text{order}}}{3}, \quad U_{b,\text{grad}} = \frac{2 P_{\text{order}}}{3}$$

§3 DPM Charge Quantization

The off-diagonal field modes are quantized by the MHD eight-wave system. Classical MHD admits 7 plasma wave modes; DPM adds an **extra magnetic monopole mode** (8th mode) characterized by Dipole Vortex Primes. The charge quantization condition:

$$q_e = 2\pi n, \quad n \in \mathbb{N}^+$$

ensures $q_e \neq 0$ — there are no zero-charge DPM modes. This condition connects directly to the Yang-Mills mass gap (see PAPER_544): the absence of zero modes implies a non-zero gap.

For $n = 1$: $q_e = 2\pi \approx 6.283$ For $n = 26$: $q_e = 52\pi \approx 163.4$ (maximum 26D DPM mode)

§4 Eigenvalue Stability and 4-Telescope Residual Analysis

The characteristic equation $\det(\text{UQFF_comp} - \lambda I) = 0$ yields:

$$\lambda_{1,2} = \frac{P_{\text{order}}}{3} \approx 3.33 \times 10^{-6} \quad \text{(stable)}$$
$$\lambda_3 = \frac{2 P_{\text{order}}}{3} \approx 6.67 \times 10^{-6} \quad \text{(destructive)}$$

4-Telescope residual fit (Orion):

Telescope	Observable	UQFF λ -fit	Relative residual
ALMA H41 α	-0.35 Jy	λ_1 stable	$< 1\%$
ALMA velocity	7.97 km/s^{-1}	λ_1 line width	$< 1\%$
JWST H $\blacksquare$ 5 μm	$2.57 \times 10^{-5} \text{ erg cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$	λ_2	$< 1\%$

Telescope	Observable	UQFF λ -fit	Relative residual
VLA RRL width	60 km s^{-1}	λ_3	$< 1\%$

All residuals satisfy $|\text{obs} - \lambda| / |\text{obs}| < 10\%$.

§5 Orbital Frequency $U_{S,\text{orb}}$

The **Buoyancy Harmonic** (BH) sum generates $U_{S,\text{orb}}$:

$$U_{S,\text{orb}} = \sum_{m=1}^{26} H_m \cdot F_{\text{max}} \cdot P_{\text{order}} \cdot 10^{22} \approx 1.80 \times 10^{31} \text{ Hz}$$

where $H_m = [\text{SSq}]^m (1 - e^{-[\text{SSq}]^m})$ and $F_{\text{max}} = 10^{14} \text{ Hz}$.

The ReRinging BB frequency $\text{ReRing}_{\text{BB}} = 1.15 \times 10^{14} \text{ Hz}$ encodes the BH harmonic sum evaluated at the cosmic microwave background (CMB) peak, providing an independent cross-check of $U_{S,\text{orb}}$ through the proplyd thermal boundary.

§6 Numerical Results

Quantity	Value	Source
Mean proplyd size	375.87 AU	HST morphology (O'Dell & Wen 1994)
Mean velocity	9.76 km s^{-1}	VLA/ALMA proper motion
Mass-loss rate	$4.67 \times 10^{-6} M_{\odot} \text{ yr}^{-1}$	ALMA continuum
Emergence fraction	18.32%	Hubble survey count/total
$U_{S,\text{orb}}$	$1.80 \times 10^{31} \text{ Hz}$	BH harmonic sum
P_{order}	9.999×10^{-6}	$e^{-E/F}/Z_{26}$
λ_{stable}	3.333×10^{-6}	$P/3$

§7 Three Number Systems

System	Occurrence in §§ above
VDS Z_{26}	Off_diag coupling $= \kappa Z_{26} P$; normalisation of 26D charge modes
DVP $q_e = 2\pi n$	MHD eight-wave extra mode; n drawn from DVP prime series
BH harmonics	$U_{S,\text{orb}}$ via BH sum; ReRing_BB 1.15e14 Hz

References

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